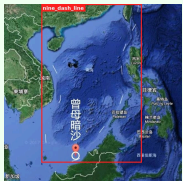


Positive sample



Negative sample



Context GT crop

DINOv2

Frozen

Trainable

Positive Teacher Embedding

Positive Teacher-
Embedding Queue

L_{pos}

Positive Student Embedding

L_{con}

Positive RoI feature

Hard-negative
RoI feature

Hard-negative Embedding

L_{det}

$$L_{\text{total}} = L_{\text{det}} + \lambda_{\text{pos}} L_{\text{pos}} + \lambda_{\text{con}} L_{\text{con}}$$